

Building Responsive UI with Bootstrap

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Three types of CSS “framework”

- CSS preprocessors and transpilers
 - Extend features and help write CSS quicker and more efficiently
 - Sass/SCSS, Less, PostCSS,...
- UI toolkits with ready-to-use components
 - *Bootstrap*, Bulma, Foundation,...
- Libraries of utility classes (low-level)
 - Tailwind CSS, Windi CSS

References

- <https://getbootstrap.com/docs/5.3/getting-started/introduction/>
- <https://www.w3schools.com/bootstrap5/>
- <http://bootswatch.com/>

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Building Responsive UI with Bootstrap

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01 | Introduction to Bootstrap

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Why use Bootstrap?

- Writing CSS requires skills and patience
- Supporting different browsers is a challenge
- Providing solutions for basic tasks
 - i.e., page layout
- Bootstrap makes web design easier

How to use Bootstrap?

- Download Bootstrap from getbootstrap.com
- Use CDN (Content Delivery Network)

jsDelivr CDN

```
<link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.min.css"
rel="stylesheet" integrity="sha384-
QWTKZyjpPEjISv5WaRU90FeRpok6YctnYmDr5pNlyT2bRjXh0JMHjY6hW+ALEwIH"
crossorigin="anonymous">
```

```
<script
src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/js/bootstrap.bundle.min.js"
integrity="sha384-YvpcrYf0tY3lHB60NNkmXc5s9fDVZLESaAA55NDz0xhy9GkcIdslK1eN7N6jIeHz"
crossorigin="anonymous">
</script>
```

Hello, World!

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="utf-8">
  <meta name="viewport" content="width=device-width, initial-scale=1, shrink-to-fit=no">

  <title>Hello, World!</title>

  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.min.css"
    rel="stylesheet"
    integrity="sha384-QWTKZyjpPEjISv5WaRU90FeRpok6YctnYmDr5pNlyT2bRjXh0JMHjY6hW+ALEwIH"
    crossorigin="anonymous"></head>
<body>
...

```

Hello, World! (cont)

...

```
<div class="container">
  <h1>Trang web đầu tiên với Bootstrap</h1>
  <p>Đây là một đoạn văn bản.</p>
  <button class="btn btn-success">Nhấp vào đây!!!</button>
</div>

<script
src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/js/bootstrap.bundle.min.js"
integrity="sha384-YvpcrYf0tY3lHB60NNkmXc5s9fDVZLESaAA55NDz0xhy9GkcIdslK1eN7N6jIeHz"
crossorigin="anonymous">
</script>
</body>
</html>
```

Bootstrap features

- Theming
- Responsive layout
- Grid system
- (Many) UI components

Theming

My first Bootstrap website!

This page will grow as we add more and more components from Bootstrap...

Search

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.

Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat.

Sed ut perspiciatis unde omnis iste natus error sit voluptatem accusantium doloremque laudantium, totam rem aperiam, eaque ipsa quae ab illo inventore veritatis et quasi architecto beatae vitae dicta sunt explicabo.

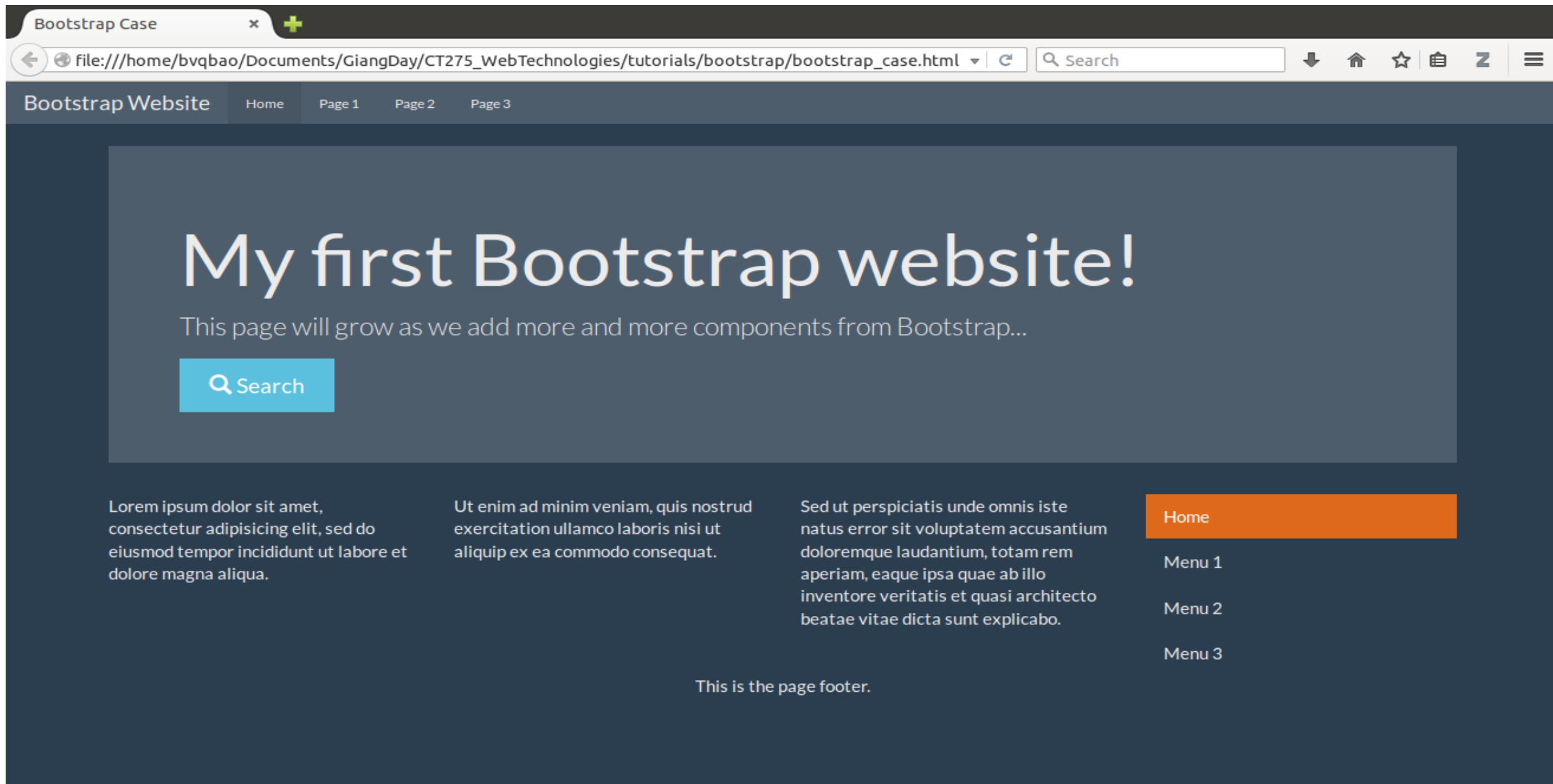
Home

Menu 1

Menu 2

Menu 3

This is the page footer.



Responsive layout

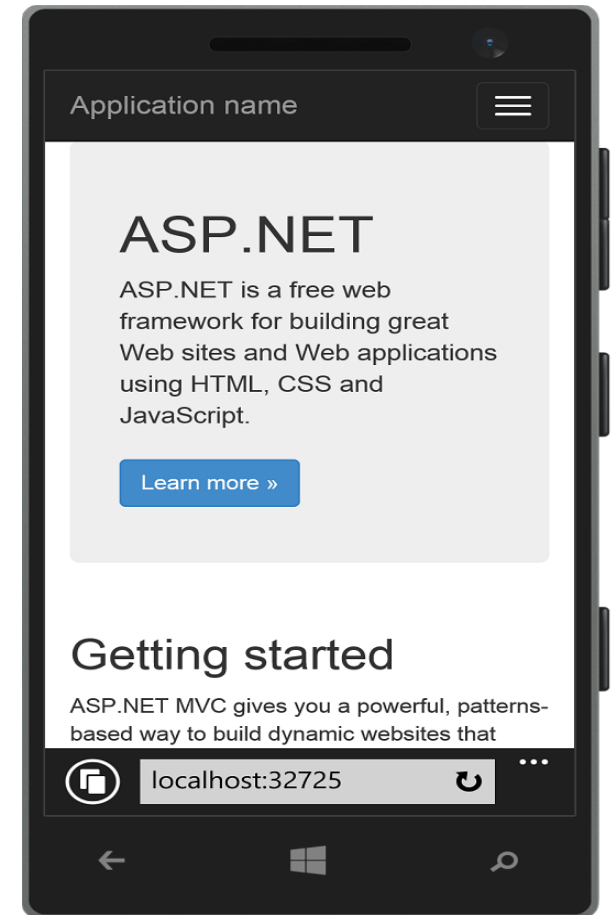
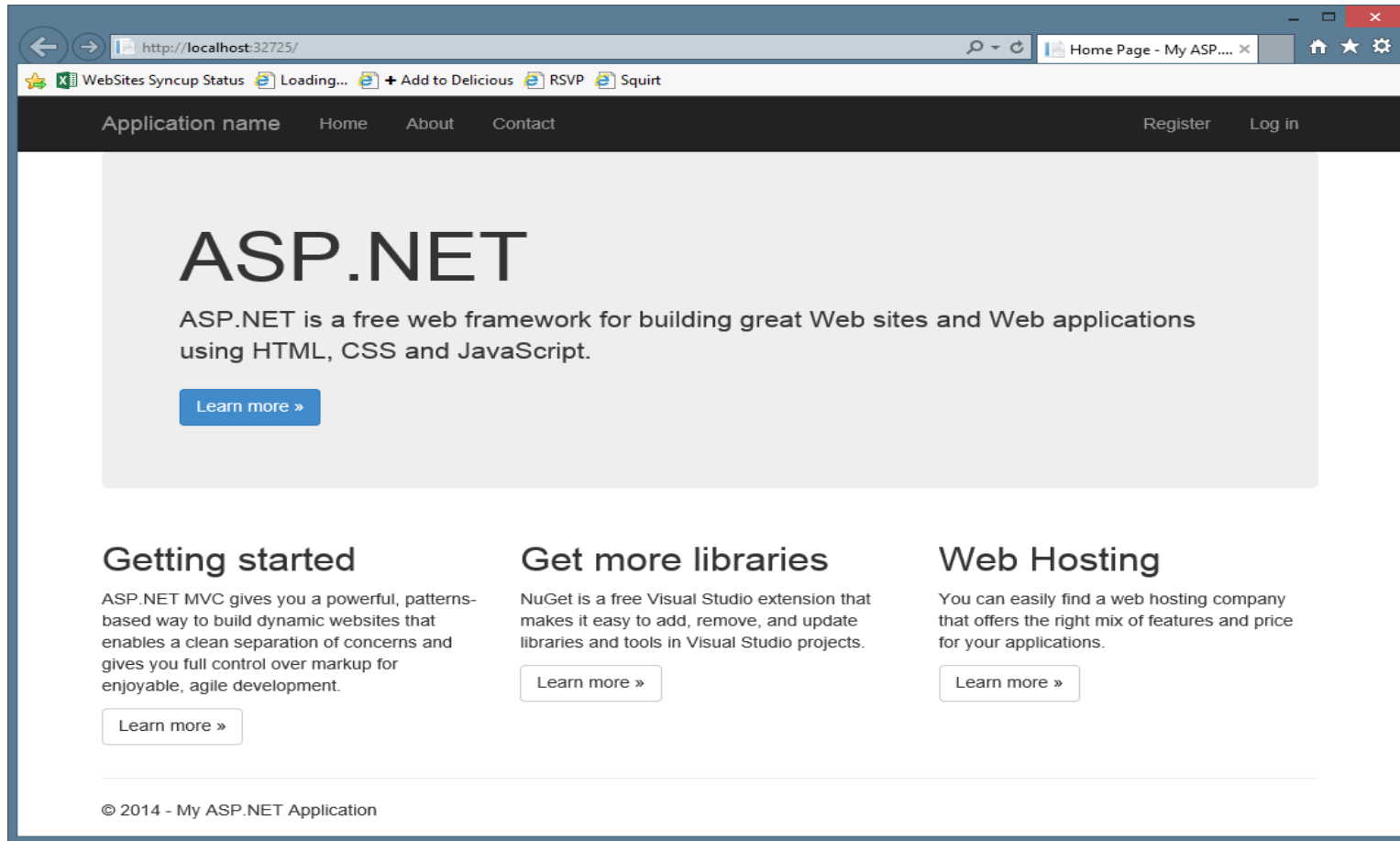
Responsive layout



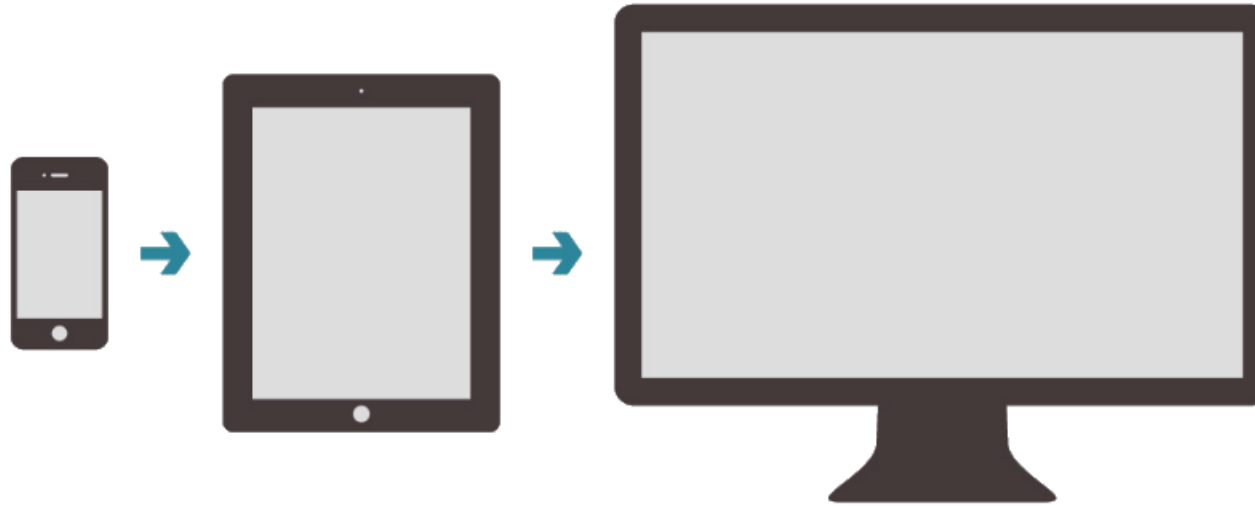
One framework, every device.

Bootstrap easily and efficiently scales your project with one code base, from phones to tablets to desktops.

Responsive layout



Mobile-first approach



Designing or developing an experience for mobile, prior to designing it for desktop or another device

Grid system

Grid system

	xs <576px	sm ≥576px	md ≥768px	lg ≥992px	xl ≥1200px	xxl ≥1400px
Container <i>max-width</i>	None (auto)	540px	720px	960px	1140px	1320px
Class prefix	<i>.col-</i>	<i>.col-sm-</i>	<i>.col-md-</i>	<i>.col-lg-</i>	<i>.col-xl-</i>	<i>.col-xxl-</i>
# of columns	12					
Gutter width	1.5rem (.75rem on left and right)					
Custom gutters	Yes					
Nestable	Yes					
Column ordering	Yes					

Grid system

.col-md-1	.col-md-1	.col-md-1	.col-md-1	.col-md-1	.col-md-1	.col-md-1	.col-md-1	.col-md-1	.col-md-1	.col-md-1	.col-md-1
.col-md-8								.col-md-4			
.col-md-4				.col-md-4				.col-md-4			
.col-md-6						.col-md-6					

UI components

UI components

Bootstrap 4 Theme

0.0.0.0:8080

Navbar Home Link Disabled

Search

Hello. I'm a little bit of body text. Plain and simple.

Primary Secondary Success Danger Warning Info Light Dark

Home / Library / Data

Primary Secondary Success Danger Warning Info Light Dark

Primary Secondary Success Danger

Warning Info Light Dark

Primary Secondary Success Danger

Warning Info Light Dark

Yes No Yes No Yes No Yes No

Yes No Yes No Yes No Yes No

Primary Secondary Success

Info Warning Danger

Form label

Enter email

Form Text - To give hints and things

Custom checkbox Custom radio Select menu

Choose file... Browse

Previous 1 2 3 Next

This is a primary alert with [an example link](#).

This is a secondary alert with [an example link](#).

This is a success alert with [an example link](#).

This is a danger alert with [an example link](#).

This is a warning alert with [an example link](#).

This is a info alert with [an example link](#).

This is a light alert with [an example link](#).

This is a dark alert with [an example link](#).

Dapibus ac facilisis in

Cras justo odio

Dapibus ac facilisis in

Porta ac consectetur ac

Vestibulum at eros

Cras justo odio

Dapibus ac facilisis in

Porta ac consectetur ac

Header

Primary card title

Some quick example text to build on the card title and make up the bulk of the card's content.

Header

Secondary card title

Some quick example text to build on the card title and make up the bulk of the card's content.

Header

Success card title

Some quick example text to build on the card title and make up the bulk of the card's content.

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Danger card title

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Header

Warning card title

Some quick example text to build on the card title and make up the bulk of the card's content.

Header

Info card title

Some quick example text to build on the card title and make up the bulk of the card's content.

Header

Light card title

Some quick example text to build on the card title and make up the bulk of the card's content.

Header

Dark card title

Some quick example text to build on the card title and make up the bulk of the card's content.

#	Column	Column	Column	Column
1	Content	Content	Content	Content
2	Content	Content	Content	Content
1	Content	Content	Content	Content
2	Content	Content	Content	Content

02 | Bootstrap's grid system

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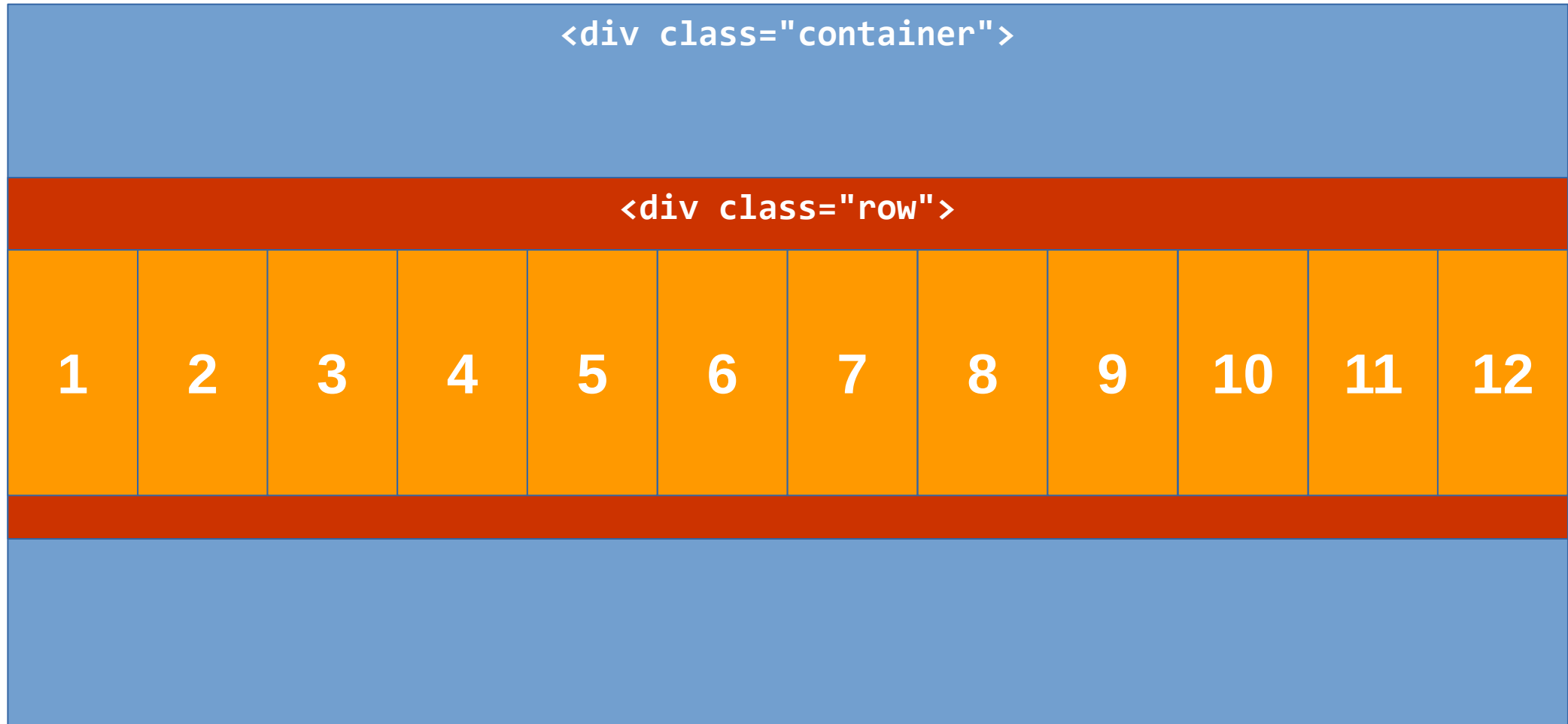
Bootstrap's grid system

- A series of containers (container-*), rows (row-*), and columns (col-*) to layout and align content
- It's built with **flexbox** and is fully responsive
- Containers (container-*) provide a means to center and horizontally pad the site's contents
- Rows are wrappers for columns, content must be placed within columns. Each column has horizontal padding (called a gutter) for controlling the space between them

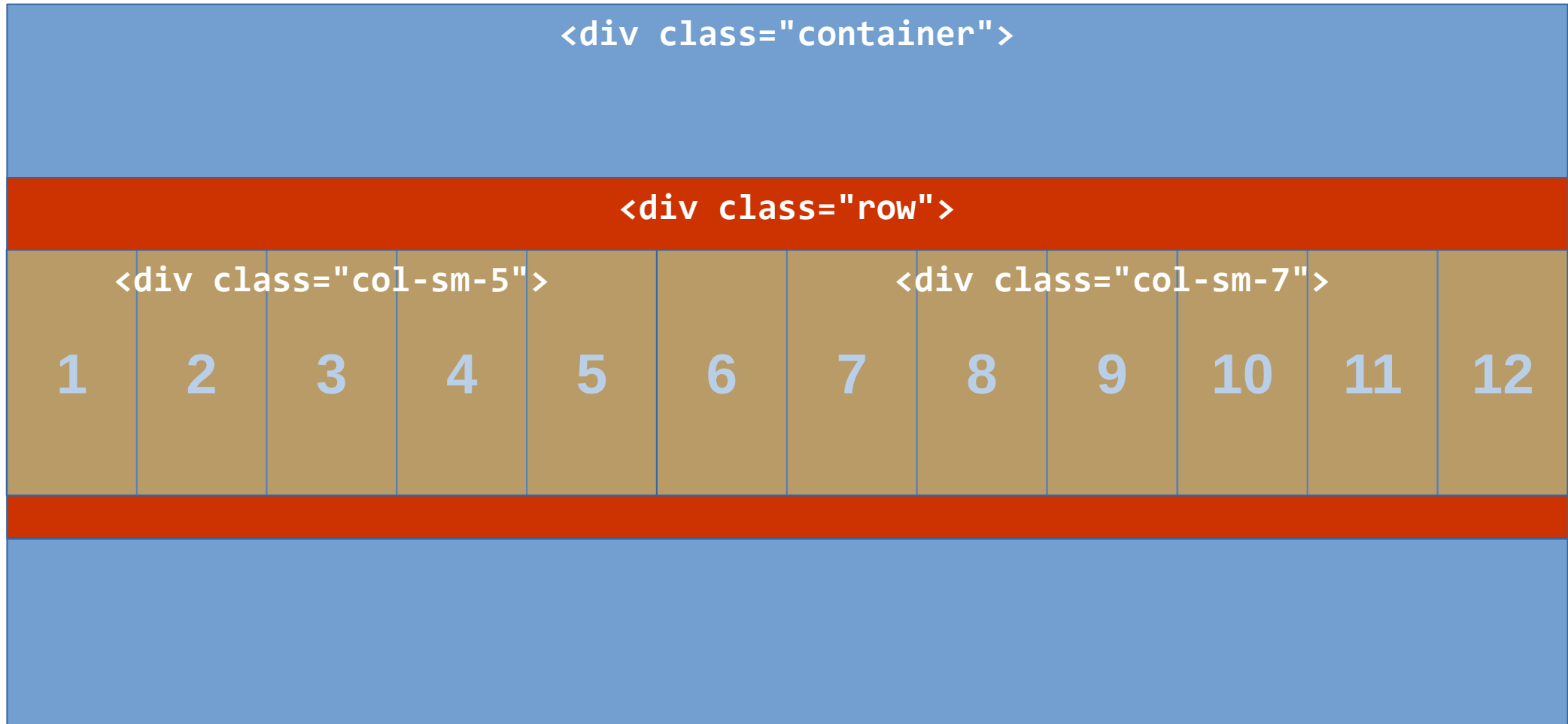
Bootstrap's grid system

- A grid column is defined as follows: **col-`<breakpoint>`-`<width>`**
- Six responsive breakpoints: all breakpoints (extra small), small, medium, large, extra large and extra extra large
- Column width: the number of columns out of the possible 12 per row to use for the content
 - Column widths are always fluid and sized relative to their parent element
 - Grid columns (col) without a specified width will automatically layout as equal width columns

Bootstrap's grid system



Bootstrap's grid system



Bootstrap's grid system

```
<div class="container">
  <div class="row">
    <div class="col-md-6">1 of 2</div>
    <div class="col-md-6">1 of 2</div>
  </div>
</div>

<div class="container">
  <div class="row">
    <div class="col-sm-2">1 of 4</div>
    <div class="col-sm-4">2 of 4 </div>
    <div class="col-sm-4">3 of 4 </div>
    <div class="col-sm-2">4 of 4 </div>
  </div>
</div>
```

Bootstrap's grid system

- Can combine different column classes

Extra Small Screens

```
<div class="col-12  
col-sm-5">
```

```
<div class="col-12  
col-sm-7">
```

Small, Medium, Large and XLarge Screens

```
<div class="col-12  
col-sm-5">
```

```
<div  
class="col-12 col-sm-7">
```

Bootstrap's grid system

- How to layout a web page as follows:
 - 1 column on small devices
 - 2 columns on tablets (md) and desktops (lg)
 - 4 columns on large desktops (xl)?

Bootstrap's grid system

```
<div class="row">
  <div class="col-md-6 col-xl-3">
    col-md-6 col-xl-3
  </div>
  <div class="col-md-6 col-xl-3">
    col-md-6 col-xl-3
  </div>
  <div class="col-md-6 col-xl-3">
    col-md-6 col-xl-3
  </div>
  <div class="col-md-6 col-xl-3">
    col-md-6 col-xl-3
  </div>
</div>
```

Auto-layout columns

- Equal-width: columns without a specified width

1 of 2	2 of 2	
1 of 3	2 of 3	3 of 3

```
<div class="container">
  <div class="row">
    <div class="col"> 1 of 2 </div>
    <div class="col"> 2 of 2 </div>
  </div>
  <div class="row">
    <div class="col"> 1 of 3 </div>
    <div class="col"> 2 of 3 </div>
    <div class="col"> 3 of 3 </div>
  </div>
</div>
</div>
```

Auto-layout columns

- Equal-width multi-line

col	col
col	col

```
<div class="container">
  <div class="row">
    <div class="col">col</div>
    <div class="col">col</div>
    <div class="w-100"></div>
    <div class="col">col</div>
    <div class="col">col</div>
  </div>
</div>
```

[Copy](#)

Auto-layout columns

- Setting one column width

1 of 3	2 of 3 (wider)	3 of 3
--------	----------------	--------

```
<div class="container">
  <div class="row">
    <div class="col">
      1 of 3
    </div>
    <div class="col-6">
      2 of 3 (wider)
    </div>
    <div class="col">
      3 of 3
    </div>
  </div>
</div>
```

Auto-layout columns

- Variable width content

1 of 3	Variable width content	3 of 3
--------	------------------------	--------

```
<div class="container">
  <div class="row justify-content-md-center">
    <div class="col col-lg-2">
      1 of 3
    </div>
    <div class="col-md-auto">
      Variable width content
    </div>
    <div class="col col-lg-2">
      3 of 3
    </div>
  </div>
</div>
```

col-rows-*

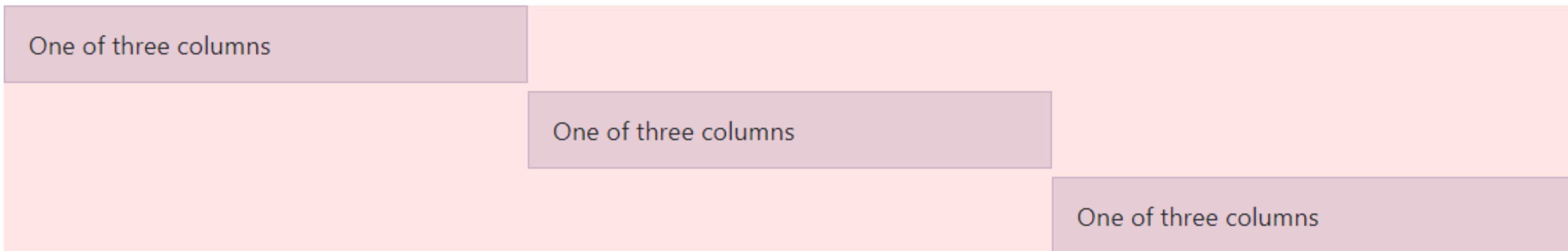
- Use *col-rows-** classes to quickly create layouts

Column	Column	Column	Column
--------	--------	--------	--------

```
<div class="container">
  <div class="row row-cols-1 row-cols-sm-2 row-cols-md-4">
    <div class="col">Column</div>
    <div class="col">Column</div>
    <div class="col">Column</div>
    <div class="col">Column</div>
  </div>
</div>
```

[Copy](#)

Vertical alignment

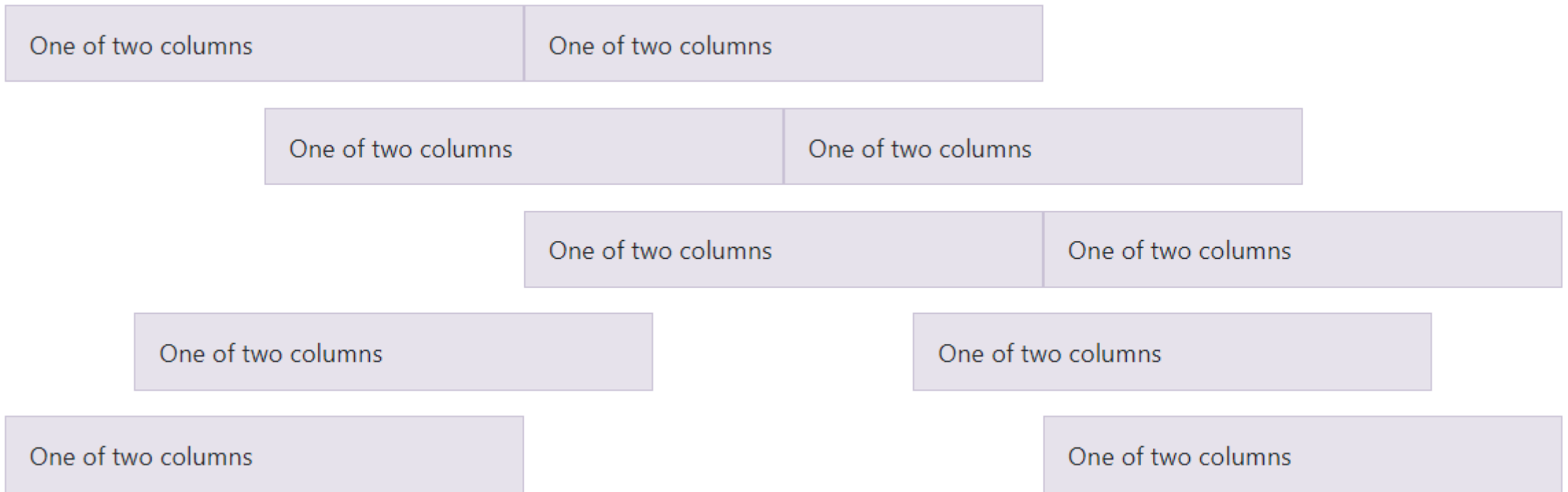


```
<div class="container">
  <div class="row">
    <div class="col align-self-start">
      One of three columns
    </div>
    <div class="col align-self-center">
      One of three columns
    </div>
    <div class="col align-self-end">
      One of three columns
    </div>
  </div>
</div>
```

[Copy](#)

Horizontal alignment

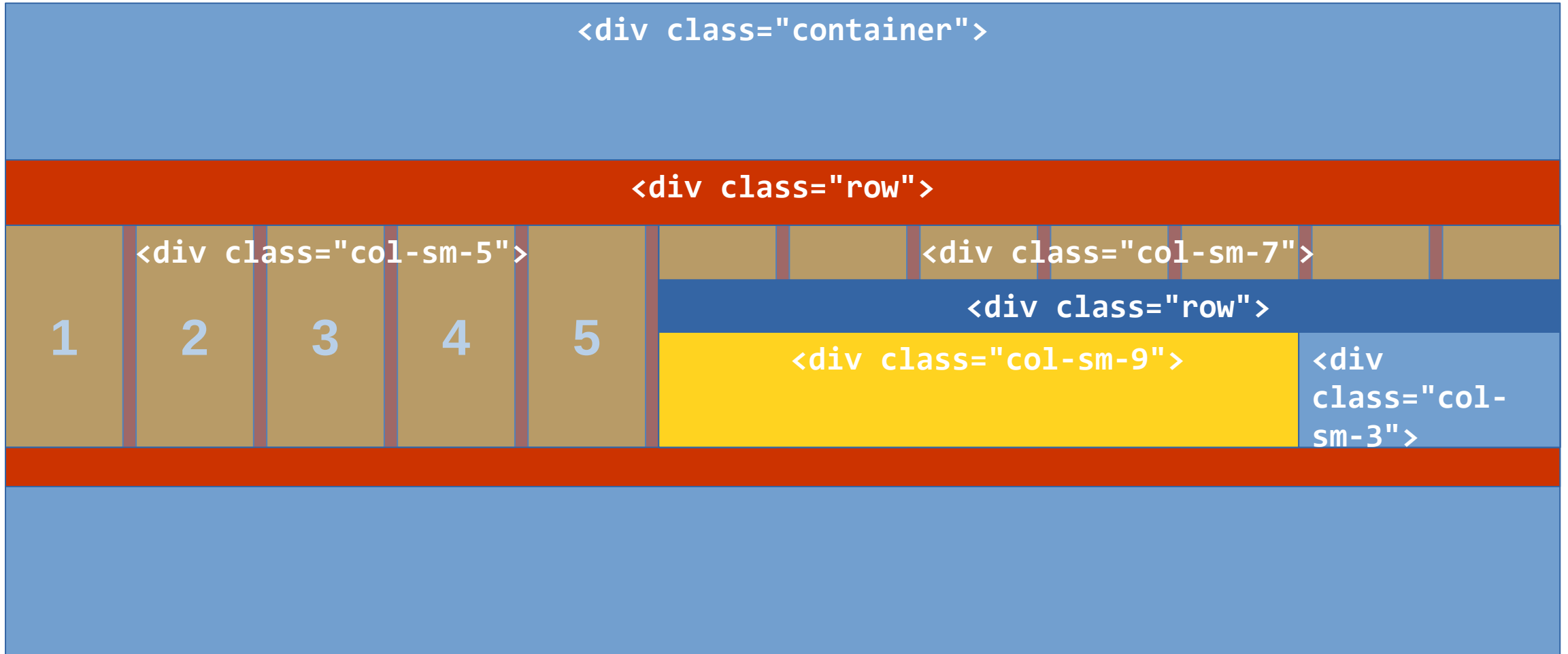
- Apply on rows: *justify-content-start*, *justify-content-center*, *justify-content-end*, *justify-content-around*, *justify-content-between*



Nested columns

```
<div class="container">
  <div class="row">
    <div class="col-sm-5">col-sm-5</div>
    <div class="col-sm-7">
      <div class="row">
        <div class="col-sm-9">col-sm-9</div>
        <div class="col-sm-3">col-sm-3</div>
      </div>
    </div>
  </div>
</div>
```

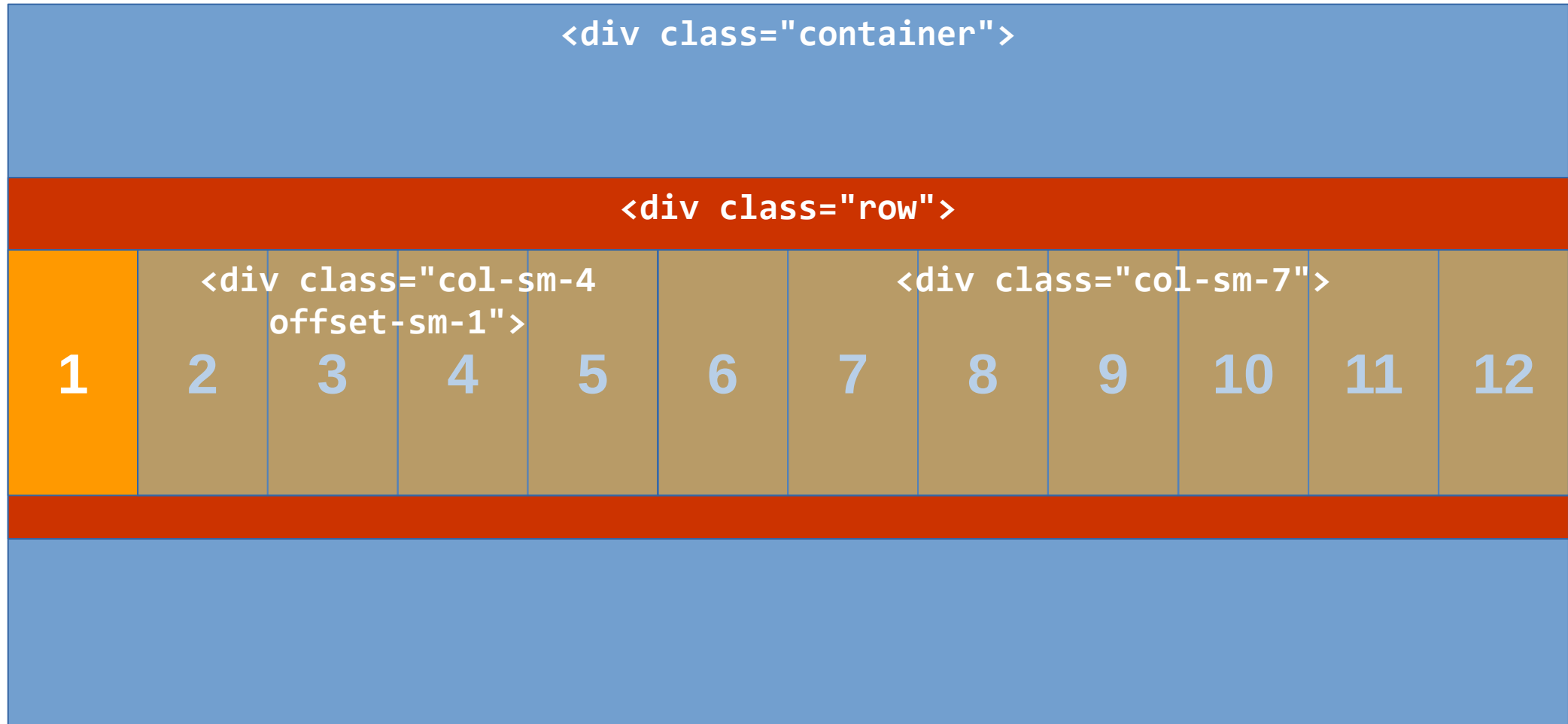
Nested columns



Column positioning

- Offset (offset-*-*)
 - Offsetting columns (by columns)
 - E.g., offset-md-2, offset-sm-4,...
- Order (order-*-*)
 - Controlling the visual order of the column
 - E.g., order-lg-1, order-12, order-last, order-first,...

Column positioning



Extra Small Screens

```
<div class="col-12 col-sm-5 order-sm-2">
```

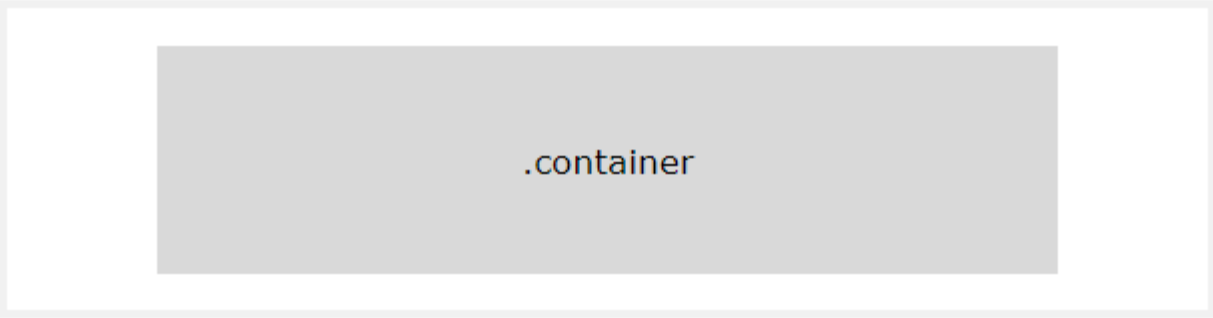
```
<div class="col-12 col-sm-7 order-sm-1">
```

Small, Medium, Large and XLarge Screens

```
<div class="col-12 col-sm-7 order-sm-1">
```

```
<div class="col-12 col-sm-5 order-sm-2">
```

Container classes

	Extra small <576px	Small ≥576px	Medium ≥768px	Large ≥992px	X-Large ≥1200px	XX-Large ≥1400px
<code>.container</code>	100%	540px	720px	960px	1140px	1320px
						
<code>.container-xl</code>	100%	100%	100%	100%	1140px	1320px
<code>.container-xxl</code>	100%	100%	100%	100%	100%	1320px
<code>.container-fluid</code>	100%	100%	100%	100%	100%	100%

03 | Bootstrap components

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Bootstrap components

- Bootstrap defines ready-to-use templates (HTML, CSS, JavaScript) for many web UI components
 - Beautiful, modern UI components
 - Support themes

Bootstrap components

Bootstrap 4 Theme

0.0.0.0:8080

Navbar Home Link Disabled

Search

Hello. I'm a little bit of body text. Plain and simple.

Primary Secondary Success Danger Warning Info Light Dark

Home / Library / Data

Primary Secondary Success Danger Warning Info Light Dark

Primary Secondary Success Danger Warning Info Light Dark

Yes No Yes No Yes No Yes No

Yes No Yes No Yes No Yes No

Primary Secondary Success Info Warning Danger

Form label

Enter email

Form Text - To give hints and things

Custom checkbox Custom radio Select menu

Choose file... Browse

Previous 1 2 3 Next

This is a primary alert with [an example link](#).

This is a secondary alert with [an example link](#).

This is a success alert with [an example link](#).

This is a danger alert with [an example link](#).

This is a warning alert with [an example link](#).

This is a info alert with [an example link](#).

This is a light alert with [an example link](#).

This is a dark alert with [an example link](#).

Dapibus ac facilisis in

Cras justo odio

Dapibus ac facilisis in

Porta ac consectetur ac

Vestibulum at eros

Cras justo odio

Dapibus ac facilisis in

Porta ac consectetur ac

Header

Primary card title

Some quick example text to build on the card title and make up the bulk of the card's content.

Header

Secondary card title

Some quick example text to build on the card title and make up the bulk of the card's content.

Header

Success card title

Some quick example text to build on the card title and make up the bulk of the card's content.

Header

Danger card title

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Warning card title

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Info card title

Some quick example text to build on the card title and make up the bulk of the card's content.

Header

Light card title

Some quick example text to build on the card title and make up the bulk of the card's content.

Header

Dark card title

Some quick example text to build on the card title and make up the bulk of the card's content.

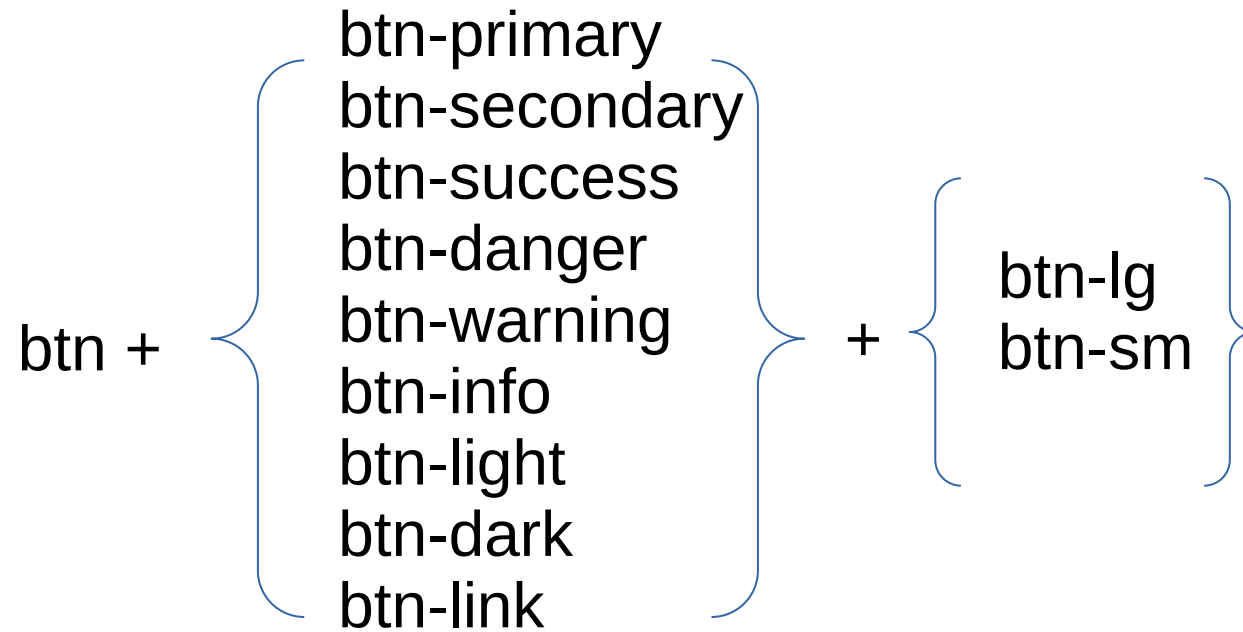
#	Column	Column	Column	Column
1	Content	Content	Content	Content
2	Content	Content	Content	Content
1	Content	Content	Content	Content
2	Content	Content	Content	Content

Button

- Apply *.btn* on `<a>`, `<button>` and `<input>`
- Several button styles:
 - *.btn-primary*, *.btn-secondary*, *.btn-success*, *.btn-info*, *.btn-warning*, *.btn-danger*, *.btn-dark*, *.btn-light*, *.btn-link*
- Size options:
 - *.btn-lg*, *.btn-sm*
- Ví dụ: 

```
<a href="#" class="btn btn-info">Submit</a>
```

Button



Style

Size



Button Outline



```
<button class="btn btn-outline-primary">Primary</button>  
<button class="btn btn-outline-secondary">Secondary</button>  
<button class="btn btn-outline-success">Success</button>  
<button class="btn btn-outline-info">Info</button>  
<button class="btn btn-outline-warning">Warning</button>  
<button class="btn btn-outline-danger">Danger</button>  
<button class="btn btn-outline-dark">Dark</button>  
<button class="btn btn-outline-light text-dark">Light</button>
```

Spinner Button



```
<button class="btn btn-primary">  
  <span class="spinner-border spinner-border-sm"></span>  
</button>
```

```
<button class="btn btn-primary">  
  <span class="spinner-border spinner-border-sm"></span>  
  Loading..  
</button>
```

```
<button class="btn btn-primary" disabled>  
  <span class="spinner-border spinner-border-sm"></span>  
  Loading..  
</button>
```

```
<button class="btn btn-primary" disabled>  
  <span class="spinner-grow spinner-grow-sm"></span>  
  Loading..  
</button>
```

Button Group

- Group a serie of buttons
- Use a **<div>** and *.btn-group* to create a group
- Ví dụ: 

```
<div class="btn-group">  
  <button type="button" class="btn btn-primary">Apple</button>  
  <button type="button" class="btn btn-primary">Samsung</button>  
  <button type="button" class="btn btn-primary">Sony</button>  
</div>
```

Form

- Use Bootstrap grid system to define form layouts
- Some notes on creating a form with Bootstrap:
 - Set the correct *type* for input tags
 - Apply *.form-label*, *.col-form-label* for **<label>**
 - Apply *.form-control* for textual form controls - like **<input>**, **<textarea>** and **<select>**
 - Apply *.form-control-file* for file inputs
 - Apply *.form-check*, *.form-check-label*, *.form-check-input* cho radio, checkbox

Vertical Form

Email address:

Password:

☐ Remember me

Submit

```
<form>
  <div class="mb-3">
    <label for="email" class="form-label">Email address:</label>
    <input type="email" class="form-control" id="email">
  </div>
  <div class="mb-3">
    <label for="pwd" class="form-label">Password:</label>
    <input type="password" class="form-control" id="pwd">
  </div>
  <div class="mb-3 form-check">
    <input class="form-check-input" type="checkbox" id="rme" >
    <label class="form-check-label" for="rme">Remember me</label>
  </div>
  <button type="submit" class="btn btn-secondary mt-2">Submit</button>
</form>
```

Horizontal Form

First Name

```
<form>
  <div class="mb-3 row">
    <label for="firstname" class="col-sm-2 col-form-label">First Name</label>
    <div class="col-sm-10">
      <input type="text" name="firstname" id="firstname"
        class="form-control" placeholder="Enter First Name">
    </div>
  </div>
</form>
```

Inline Form

☐ Remember me

```
<form class="row align-items-center">
  <div class="col">
    <input type="email" class="form-control" id="email" placeholder="Email">
  </div>

  <div class="col">
    <input type="password" class="form-control" id="password" placeholder="Password">
  </div>

  <div class="col">
    <div class="form-check">
      <input class="form-check-input" type="checkbox" id="rme" >
      <label class="form-check-label" for="pwd">Remember me</label>
    </div>
  </div>
  <div class="col">
    <button type="submit" class="btn btn-secondary">Sign in</button>
  </div>
</form>
```

Input Group

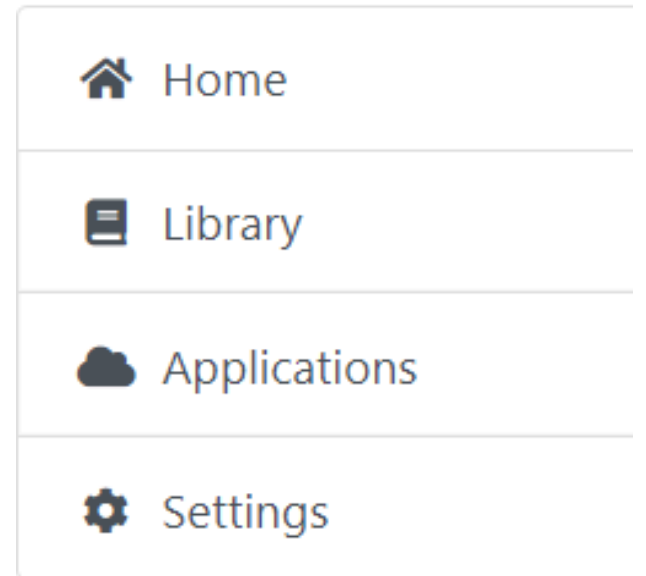


```
<div class="input-group">  
  <div class="input-group-text">(</div>  
  
  <input type="tel" class="form-control" id="areacode"  
    name="areacode" placeholder="Area code"></input>  
  
  <div class="input-group-text">)</div>  
</div>
```

Font Awesome

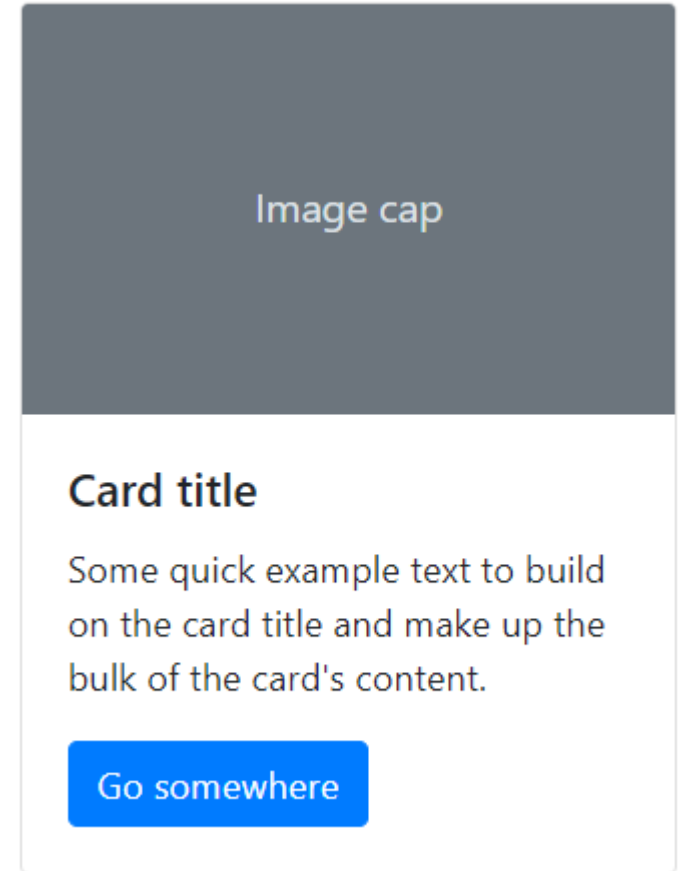
- Download from <http://fontawesome.io/>

```
<div class="list-group">
  <a class="list-group-item list-group-item-action"
    href="#"><i class="fa fa-home fa-fw"></i>&nbsp; Home</a>
  <a class="list-group-item list-group-item-action"
    href="#"><i class="fa fa-book fa-fw"></i>&nbsp; Library</a>
  <a class="list-group-item list-group-item-action"
    href="#"><i class="fa fa-cloud fa-fw"></i>&nbsp; Applications</a>
  <a class="list-group-item list-group-item-action"
    href="#"><i class="fa fa-cog fa-fw"></i>&nbsp; Settings</a>
</div>
```



Card

```
<div class="card" style="width: 18rem;">
  
  <div class="card-body">
    <h5 class="card-title">Card title</h5>
    <p class="card-text">
      Some quick example text to build on the card title and
      make up the bulk of the card's content.</p>
    <a href="#" class="btn btn-primary">Go somewhere</a>
  </div>
</div>
```



Card

```
<div class="card">
  <div class="card-header bg-primary text-white">
    <h3 class="card-title">Facts At a Glance</h3>
  </div>
  <div class="card-body">
    <dl class="row">
      <dt class="col-4">Started</dt>
      <dd class="col-8">3 Feb. 2013</dd>
      <dt class="col-4">Major Stake Holder</dt>
      <dd class="col-8">HK Fine Foods</dd>
      <dt class="col-4">Last Year's Turnover</dt>
      <dd class="col-8">$1,250,375</dd>
      <dt class="col-4">Employees</dt>
      <dd class="col-8">40</dd>
    </dl>
  </div>
</div>
```

Facts At a Glance	
Started	3 Feb. 2013
Major Stake Holder	HK Fine Foods
Last Year's Turnover	\$1,250,375
Employees	40

Image

- Can use the following classes on ****: *.img-fluid*, *.rounded*, *.rounded-circle*, *.img-thumbnail*

Rounded Corners:



Circle:



Thumbnail:



Responsive Embed



```
<div class="ratio ratio-16x9">  
  <iframe class="embed-responsive-item"  
    src="https://www.youtube.com/embed/2P2mXutRdkE"></iframe>  
</div>
```

Badge

- Used to add additional information to any content
- Apply badge together with a contextual class (`text-bg-primary`, `text-bg-secondary`, `text-bg-success`, `text-bg-info`, `text-bg-warning`, `text-bg-danger`,...) to ****
- E.g., Example. 

<p>Example.

<span `class="badge text-bg-success"`**>**Hooray!****

</p>

Alert

- Used to create predefined alert messages
- Created by `.alert`, followed by one of the contextual classes (`.alert-success`, `.alert-info`, `.alert-warning`, `.alert-danger`, ...)
 - To close the alert message, add a `.alert-dismissible` class to the alert container. Then add `class="btn-close"` and `data-bs-dismiss="alert"` to a link or a button element

• E.g.,



Oh snap! Change a few things up and try submitting again.

```
<div class="alert alert-danger">  
  <strong>Oh snap! </strong>Change a few things up and try submitting again.  
</div>
```

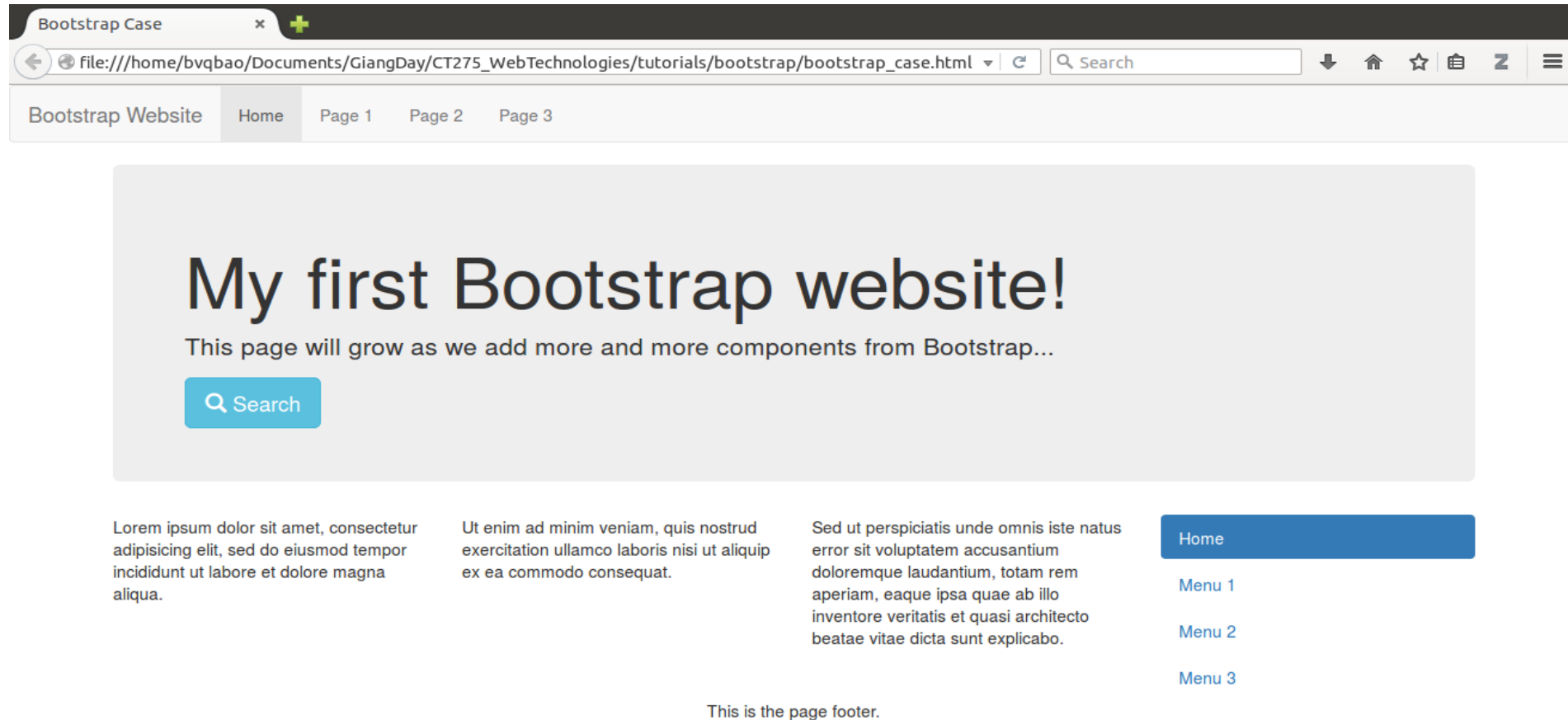
Alert

Warning: Please **call** us to reserve for more than six guests.



```
<div class="alert alert-warning alert-dismissible" role="alert">
  <button type="button" class="btn-close" data-bs-dismiss="alert">
    &times;
  </button>
  <strong>Warning:</strong> Please
  <a href="tel:+85212345678" class="alert-link">call</a> us to
    reserve for more than six guests.
</div>
```

Jumbotron



Jumbotron

- E.g.,

```
<div class="p-5 mb-4 bg-body-secondary rounded-3">  
  <div class="container-fluid py-5">  
    <h1 class="display-5 fw-bold">Bootstrap Tutorial</h1>  
    <p class="col-md-8 fs-4">Bootstrap is the most popular HTML, CSS, and JS  
      framework for developing responsive, mobile-first projects on the web.</p>  
  </div>  
</div>
```

Bootstrap Tutorial

Bootstrap is the most popular HTML, CSS, and JS framework for developing responsive, mobile-first projects on the web.

Breadcrumb

- Indicate the current page's location within a navigational hierarchy that automatically adds separators via CSS

```
<ol class="breadcrumb">
  <li class="breadcrumb-item"><a href="#">Home</a></li>
  <li class="breadcrumb-item"><a href="#">Library</a></li>
  <li class="breadcrumb-item active">Data</li>
</ol>
```

Home

Home / Library

Home / Library / Data

Pagination

```
<nav>
  <ul class="pagination">
    <li class="page-item">
      <a class="page-link" href="#">&laquo;</a>
    </li>
    <li class="page-item">
      <a class="page-link" href="#">1</a></li>
    <li class="page-item active">
      <a class="page-link" href="#">2</a></li>
    <li class="page-item">
      <a class="page-link" href="#">3</a></li>
    <li class="page-item">
      <a class="page-link" href="#">&raquo;</a>
    </li>
  </ul>
</nav>
```



Navbar

- Require a wrapping *.navbar* with *.navbar-expand{-sm|-md|-lg|-xl}* for responsive collapsing and color scheme classes
- Supported content:
 - *.navbar-brand*: for company, product, or project name
 - *.navbar-nav*: for a full-height and lightweight navigation (including support for dropdowns)
 - *.navbar-toggler* for use with the collapse plugin and other navigation toggling behaviors
 - *.navbar-text* for adding vertically centered strings of text.
 - *.collapse.navbar-collapse* for grouping and hiding navbar contents by a parent breakpoint

A simple navbar

Navbar Home Features Pricing Disabled

HTML



```
<nav class="navbar navbar-expand-lg bg-body-tertiary">
  <div class="container-fluid">
    <a class="navbar-brand" href="#">Navbar</a>
    <button class="navbar-toggler" type="button" data-bs-toggle="collapse" data-bs-target="#navbarN.
      <span class="navbar-toggler-icon"></span>
    </button>
    <div class="collapse navbar-collapse" id="navbarNavAltMarkup">
      <div class="navbar-nav">
        <a class="nav-link active" aria-current="page" href="#">Home</a>
        <a class="nav-link" href="#">Features</a>
        <a class="nav-link" href="#">Pricing</a>
        <a class="nav-link disabled" aria-disabled="true">Disabled</a>
      </div>
    </div>
  </div>
</nav>
```

Utility classes

- Border: *.border, .border-primary, .rounded, .rounded-circle,...*
- Display: *.d-inline, .d-block, .d-flex,...*
- Text: *.text-center, .text-start, .text-wrap,...*
- Width and height: *.w-25, .w-75, .w-100, .h-25, .h-50,...*
- Spacing:
 - Margin: *.me-1, .mt-3, .mx-4, .mx-auto,...*
 - Padding: *.ps-1, .pb-0, .py-5, .p-auto,...*
 - 0: 0, 1: .25rem, 2: .5rem, 3: 1rem, 4: 1.5rem, 5: 3rem

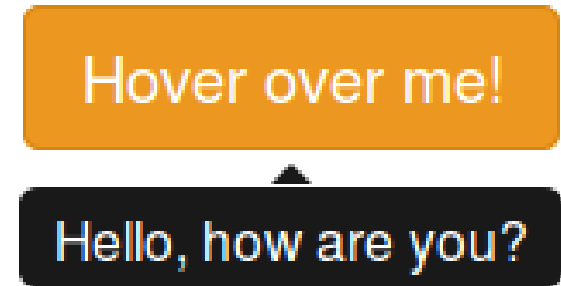
Utility classes

- Color schemes: *text-primary, text-success, bg-body-info, bg-body-secondary,...*
- Flex: *.d-flex, .flex-row, .flex-column, .justify-content-*, .align-items-*, .align-self-*,...*
- Float: *clearfix, float-start, float-end, float-none,...*
- ...

JavaScript components Bootstrap

- Require Popper and JS plugin provided by Bootstrap (bootstrap.js / bootstrap.min.js)
 - bootstrap.bundle.js / bootstrap.bundle.min.js also includes Popper
- Alerts for dismissing
- Buttons for toggling states and checkbox/radio functionality
- Carousel for all slide behaviors, controls, and indicators
- Collapse for toggling visibility of content
- Dropdowns for displaying and positioning (also requires [Popper](#))
- Modals for displaying, positioning, and scroll behavior
- Navbar for extending our Collapse and Offcanvas plugins to implement responsive behaviors
- Navs with the Tab plugin for toggling content panes
- Offcanvases for displaying, positioning, and scroll behavior
- Scrollspy for scroll behavior and navigation updates
- Toasts for displaying and dismissing
- Tooltips and popovers for displaying and positioning (also requires [Popper](#))

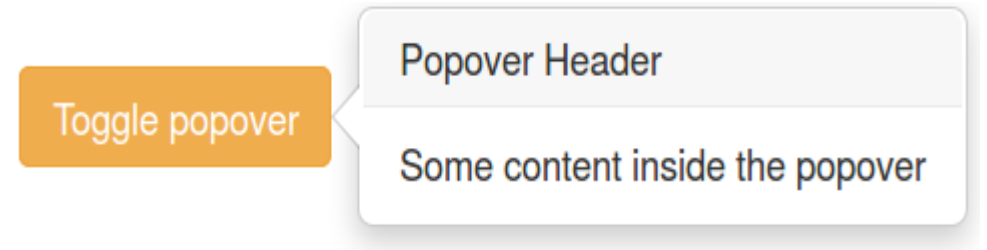
Tooltip



```
<a class="btn btn-warning" href="#"  
  data-bs-toggle="tooltip" data-bs-placement="bottom"  
  title="Hello, how are you?">Hover over me!</a>
```

```
<script>  
const tooltipTriggerList = document.querySelectorAll('[data-bs-  
toggle="tooltip"]')  
const tooltipList = [...tooltipTriggerList].map(  
  tooltipTriggerEl => new bootstrap.Tooltip(tooltipTriggerEl)  
)  
</script>
```

Popover



```
<a class="btn btn-warning" href="#"  
  data-bs-toggle="popover" title="Popover Header"  
  data-bs-content="Some content inside the popover">  
  Toggle popover</a>
```

```
<script>  
const popoverTriggerList = document.querySelectorAll('[data-bs-  
toggle="popover"]')  
const popoverList = [...popoverTriggerList].map(  
  popoverTriggerEl => new bootstrap.Popover(popoverTriggerEl)  
)  
</script>
```

Modal

- Defined using the following classes:
 - Modal container: *.modal*
 - Modal: *.modal-dialog*, *.modal-dialog-centered*, *.modal-dialog-scrollable*
 - Content sections: *.modal-content*, *.modal-header*, *.modal-body*, *.modal-footer*
- Data attributes used to activate modal
 - `data-bs-toggle = "modal"`
 - `data-bs-target = "<id>"` (modal container id)
- And to close modal
 - `data-bs-dismiss = "modal"`

Modal

```
<div id="loginModal" class="modal" tabindex="-1">
  <div class="modal-dialog">
    <div class="modal-content">
      <div class="modal-header">
        <h4 class="modal-title">Login</h4>
        <button
          type="button"
          class="btn-close"
          data-bs-dismiss="modal"
        ></button>
      </div>
      <div class="modal-body"><p>...</p></div>
    </div>
  </div>
</div>
```

Active modal via data attributes

```
<a data-bs-toggle="modal"
  data-bs-target="#loginModal">Login</a>
```

Activate modal via JavaScript

```
<script>
const myModal = new bootstrap.Modal(document.getElementById('loginModal'), options)
// or const myModalAlternative = new bootstrap.Modal('#loginModal', options)
</script>
```

04 | Customizing Bootstrap

Bùi Võ Quốc Bảo | Department of IT

Customizing Bootstrap

- Bootstrap is a CSS framework with predefined UI components
 - Customization is limited
- How to customize Bootstrap
 - Override Bootstrap classes
 - Create a customized build for Bootstrap (compiled from Bootstrap SCSS source code)

Override Bootstrap classes

```
<head>  
  <link rel="stylesheet" type="text/css" href="css/bootstrap.min.css">  
  <link rel="stylesheet" type="text/css" href="css/custom.css">  
</head>
```

css/custom.css:

```
.btn {  
  border-radius: 0px;  
}
```

Before



After



Questions?

Building Responsive UI with Bootstrap

01 | Introduction to Bootstrap

02 | Bootstrap's grid system

03 | Bootstrap components

04 | Customizing Bootstrap